

6th Grade Written Exam

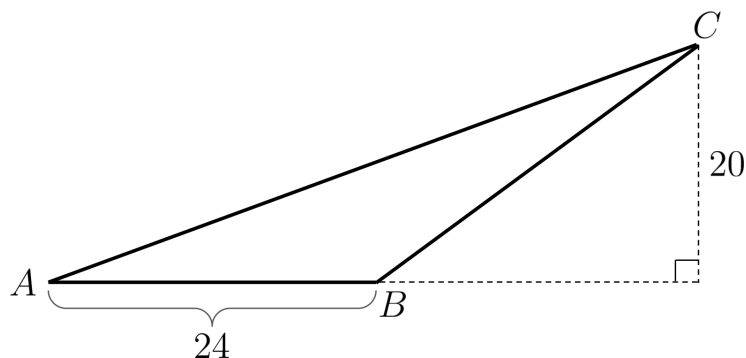
2026 James Clemens Math Tournament

INSTRUCTIONS:

1. You will have 90 minutes to complete this examination.
2. This exam consists of 25 multiple choice questions and 3 tiebreaker questions. The multiple choice questions are each worth 4 points if answered correctly and -1 points if answered incorrectly. There is no penalty for unanswered questions. The tiebreaker questions are worth 0.1 points if answered correctly and 0 points if answered incorrectly or left unanswered.
3. Use a #2 pencil.
4. Calculators, cellular devices, books, notes, and other aids are prohibited during this exam. Scratch paper will be provided for calculations. If you require additional scratch paper, simply raise your hand and your proctor will assist you.
5. Mark your answers to the questions in the Scantron form provided. You may use the test booklets for scratch work, but only the answers marked in the Scantron form will be counted. Make sure to fill in the bubbles on the Scantron form **completely** and erase any stray marks.
6. Write your answers to the tiebreaker questions on the back of the Scantron form. Label the tiebreakers as TB1, TB2, and TB3.
7. NOTA denotes "None Of The Above."
8. Diagrams are not necessarily drawn to scale.
9. In the event of a tie, answers will be evaluated backwards starting from multiple choice question 25 to determine a winner.
10. Although this math tournament is intended to demonstrate your knowledge and skills in math, be sure to enjoy the experience and have fun!
11. You may keep the test booklet after the exam.

DO NOT TURN THIS PAGE UNTIL TOLD TO DO SO

- Compute: $(1 + 2 + 3 + \cdots + 10) - (2 + 4 + 6 + 8 + 10)$
 (A) 15 (B) 25 (C) 30 (D) 55 (E) NOTA
- Peyton is waiting in a straight line of people to get lunch. He counts 6 people in front of him and 25 people behind him. How many people are in the line?
 (A) 29 (B) 30 (C) 31 (D) 32 (E) NOTA
- Molly makes muffins on Mondays from 9:00 am to 11:00 am at a rate of 5 muffins every twenty minutes. If she has to make 300 muffins for Milo, she calculates it will take x Mondays to prepare. What is the value of x ?
 (A) 1 (B) 5 (C) 10 (D) 20 (E) NOTA
- Find the area of triangle ABC .

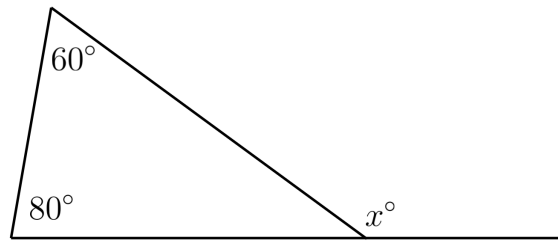


- (A) 44 (B) 240 (C) 480 (D) 960 (E) NOTA
- The table below shows Morgan's daily screen time for the past week, in minutes. What is the median number of minutes she spent on her screen per day?

	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Time (min)	105	55	47	82	63	78	130

 (A) 25 (B) 63 (C) 78 (D) 80 (E) NOTA
- A water faucet is filling a 120-liter bathtub with water at a rate of 10 liters per minute. However, the drain is not completely closed, and water is draining at a rate of 4 liters per minute. How many minutes will it take to fill the bathtub?
 (A) 12 (B) 20 (C) 24 (D) 30 (E) NOTA

7. To get to the grocery store from his house, Albert must travel 4 miles north and 3 miles east. What is the straight-line distance between Albert's house and the store?
 (A) 3 (B) 4 (C) 5 (D) 7 (E) NOTA
8. Samantha and her friends watched the movie "Kpop Demon Hunters" over the weekend. On Saturday, they watched the first $\frac{1}{3}$ of the movie. On Sunday, they watched the remaining 1 hour. How long is the movie, in minutes?
 (A) 75 (B) 90 (C) 120 (D) 180 (E) NOTA
9. If $a \triangle b = a^2 + b$, compute $3 \triangle (4 \triangle 5)$.
 (A) 30 (B) 44 (C) 174 (D) 844 (E) NOTA
10. Jasmine runs at a speed of 6 mph on flat ground and 3 mph when running uphill. During a competition, she ran 3 miles on flat ground and 1 mile uphill. How long did she run, in minutes?
 (A) $\frac{5}{6}$ (B) 5 (C) 50 (D) 300 (E) NOTA
11. Simplify: $(4 + 8 \div 4) \cdot (1^{2026} - (4 - 7) + 5 \cdot 4)$
 (A) 30 (B) 60 (C) 72 (D) 144 (E) NOTA
12. Find the value of x in the diagram.



- (A) 120 (B) 130 (C) 140 (D) 150 (E) NOTA
13. Simplify: $\sqrt{144} + 2^3 - \sqrt[3]{27}$
 (A) 17 (B) 20 (C) 23 (D) 29 (E) NOTA
14. A $51\frac{2}{7}$ -inch strip of wood is cut into two sections. If one section is $27\frac{6}{7}$ inches long, how long is the other section, in inches?
 (A) $23\frac{2}{7}$ (B) $23\frac{3}{7}$ (C) $23\frac{4}{7}$ (D) $23\frac{5}{7}$ (E) NOTA

15. Let A equal the greatest common factor of 52 and 91, and let B be the number of prime numbers between 1 and 25. What is $A + B$?
- (A) 10 (B) 11 (C) 16 (D) 17 (E) NOTA
16. Each side of a 4×5 rectangle is increased by 1 unit. By what percent does the area of the rectangle increase?
- (A) 25% (B) 50% (C) 75% (D) 150% (E) NOTA
17. Austin has two bags of coins. The first bag has 3 red coins and 3 blue coins, and the second bag has 6 green coins and 2 purple coins. He randomly takes out a coin from each of the two bags. What is the probability that he takes out a red coin from the first bag and a purple coin from the second?
- (A) $\frac{1}{8}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$ (E) NOTA
18. How many positive integers below 1000 are divisible by 18 or 43?
- (A) 23 (B) 55 (C) 77 (D) 78 (E) NOTA
19. Let a , b , and c be real numbers that satisfy

$$\begin{aligned}2a + b + c &= 47 \\a + 2b + c &= 159 \\a + b + 2c &= 174.\end{aligned}$$

What is $a + b + c$?

- (A) 95 (B) 128 (C) 196 (D) 384 (E) NOTA
20. Let x be a 2-digit number. Let y be the sum of the digits of x . The number $x + y$ contains the same digits as x but in reverse order. Find y .
- (A) 6 (B) 7 (C) 8 (D) 9 (E) NOTA
21. A square has three of its coordinates on the points $(-3, 1)$, $(5, 1)$, and $(5, -7)$. What is the sum of the x and y -coordinates of the center of the square?
- (A) -10 (B) -2 (C) 2 (D) 4 (E) NOTA

Tiebreakers:

TB1. How many times does the letter “m” appear in Problem 3?

TB2. Compute

$$1 + \frac{(\sum_{k=2}^{2026} \lfloor \log_k 67k \rfloor) \left(\frac{7^2 - 2 \cdot 5^2 + 1}{2026} \right)}{\sqrt{e + \sqrt{e + \sqrt{e + \dots}}}} \cdot \int_0^\pi e^{2x} \sec e^{2x} \tan e^{2x} dx$$

Note that $e \approx 2.718$, and $\lfloor a \rfloor$ denotes the greatest integer less than or equal to a .

TB3. Dr. Al Gebra is driving on a highway with speed limit 60 mph. There are two police cars parked on the shoulder of the highway at two points. When he passes the first police car, his speed is 50 mph. Four minutes and four miles later, when he passes the second police car, his speed is also 50 mph. Should the second policeman chase down Dr. Al Gebra for speeding? Answer with “yes” or “no”. (Assume that Dr. Al Gebra is considered speeding if at any point between the two police cars his speed exceeds 60 mph.)